

SecureGate: Gate Interface and Initial Gate-Fingerprint Analysis

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Purpose. This report documents the work implemented in `02_gate_interface.ipynb`. The notebook establishes the logical-gate interface used by SecureGate, audits the available surface-code backend, implements encoded logical X and Z operations, samples their detector records, and performs an initial gate-inference test. Its purpose is to provide a compact scientific roadmap so that the notebook can be followed in execution order without reconstructing the motivation of each cell.

1. Logical-gate interface.

A common `GateOperation` representation was introduced for logical operations. The interface stores a gate name and the logical-qubit indices on which the gate acts. The current SecureGate gate set is

$$\{I, X, Y, Z, H, S, CX, T\}.$$

The implementation normalizes gate names to uppercase, checks that the gate belongs to the supported set, enforces the expected arity (one logical qubit for single-qubit gates and two for CX), and requires non-negative integer qubit indices. Basic tests verified valid X and CX objects and confirmed that an invalid one-qubit CX operation is rejected. These tests establish a consistent front-end representation before fault-tolerant implementations are attached.

2. Surface-code backend audit.

The installed `surface-sim` interface was inspected rather than assuming that every required logical gate is natively available. For the rotated CSS surface-code mapping used in the experiments, the directly exposed logical operations include X , Z , S , and CX . The inspected unrotated mapping also exposes H . Y and T are not directly exposed by either mapping in the installed interface.

Accordingly, Y and T are treated as implementation tasks requiring an explicit construction (for example, a decomposition or state-injection strategy) rather than being assumed to be native operations. This capability audit is important for keeping later claims tied to the actual simulator API.

3. Distance-5 logical X implementation.

The gate-fingerprint experiments use a distance-5 rotated surface code. The constructed layout contains 49 physical qubits, comprising 25 data qubits and 24 ancilla qubits, and represents one logical qubit.

For the logical X operator, `surface-sim` identifies the physical support as

$$\{D1, D6, D11, D16, D21\},$$

which correspond to physical indices

$$\{0, 5, 10, 15, 20\}.$$

The built-in X -gate configuration was inspected and applied to the layout. The generated logical-operation block explicitly contains

$$X\ 0\ 5\ 10\ 15\ 20,$$

with the same physical error probability applied through the noise model. The implementation check returned `True`, confirming that the generated physical X layer matches the logical- X support specified by the layout.

The high-level logical circuit is initialized, advanced by a `TICK`, applies X , advances by a second `TICK`, and finally measures the logical qubit. After encoding through `surface-sim`, the complete fault-tolerant circuit contains 49 physical qubits, 73 measurements, 60 detectors, one logical observable, and 27 temporal ticks. The additional gates visible in the encoded circuit are part of the syndrome-extraction procedure rather than additional logical gates.

4. Detector sampling and X_L fingerprint.

For each experiment, the detector sampler returns a binary record for every shot. A detector value $D_j = 1$ denotes a detector event at detector j , while $D_j = 0$ denotes no event. The experiment was sampled for 10,000 shots,

producing a

$$10,000 \times 60$$

binary detector matrix.

For detector j , the empirical event probability was estimated as

$$\hat{P}(D_j = 1 | X_L) = \frac{1}{N} \sum_{n=1}^N D_j^{(n)}, \quad N = 10,000.$$

The detector coordinates provide a spacetime location (x, y, t) for each detector, allowing the empirical fingerprint to be examined both detector-by-detector and by temporal layer.

The measured X_L detector activity is concentrated around the logical-operation layer. The mean detector-event probability was 0.010087 at $t = 0$, 0.035963 at $t = 1$, and 0.020167 at $t = 1.5$. The largest observed detector-event probabilities were located at $t = 1$. This establishes a time-localized detector signature associated with the logical-operation region, but it does not by itself establish unique gate identification.

5. Logical Z experiment and detector correspondence.

A separate distance-5 logical Z experiment was constructed using the same layout, detector frame, and physical error probability. The logical Z support is the corresponding string of five data qubits $D1$ – $D5$, giving physical indices 0, 1, 2, 3, 4.

The X_L and Z_L experiments were verified to contain the same detector count and the same ordered spacetime detector coordinates. Therefore, detector j in one experiment corresponds to detector j in the other experiment, which permits direct comparison of

$$\hat{P}(D_j = 1 | X_L) \quad \text{and} \quad \hat{P}(D_j = 1 | Z_L).$$

The mean absolute detector-level difference was

$$\frac{1}{60} \sum_{j=1}^{60} \left| \hat{P}(D_j = 1 | X_L) - \hat{P}(D_j = 1 | Z_L) \right| = 0.001635.$$

The maximum absolute difference was 0.00510, occurring at detector 26, with spacetime coordinate

$$(x, y, t) = (2, 4, 1).$$

At this detector,

$$\hat{P}(D_{26} = 1 | X_L) = 0.04050, \quad \hat{P}(D_{26} = 1 | Z_L) = 0.03540.$$

The temporal comparison gives

t	X_L mean	Z_L mean	$ \Delta $
0.0	0.010087	0.010271	0.000183
1.0	0.035963	0.035046	0.000917
1.5	0.020167	0.020658	0.000492

so the largest aggregate difference is also concentrated at the central logical-operation layer.

6. Initial gate-inference experiment.

The complete 60-detector record from each shot was used as the feature vector. The 10,000 X_L and 10,000 Z_L shots were combined into a balanced binary dataset. An 80/20 stratified train/test split produced 8,000 training shots and 2,000 test shots per gate.

A Bernoulli Naive Bayes classifier was trained because detector outcomes are binary. For an unseen detector record, the classifier estimates which gate class is more compatible with the observed pattern. The resulting test set contained 4,000 shots in total and gave

$$\text{Accuracy} = 0.5052.$$

The confusion matrix was

	$\hat{X}_L$	$\hat{Z}_L$
X_L	537	1463
Z_L	516	1484

showing a substantial bias toward predicting Z_L . The approximate 95% confidence interval for the measured accuracy was

$$[0.4898, 0.5207],$$

which contains the 0.50 chance level for two equally represented classes.

These results indicate that, under the present simulation conditions and for this particular Bernoulli Naive Bayes baseline, the detector patterns did not provide reliable single-shot X_L -versus- Z_L identification. This should not be interpreted as a proof that the two gate distributions are exactly identical, or that no inference method can ever distinguish them. Rather, small empirical differences in detector firing probabilities did not translate into classification performance clearly above chance.

7. Relation to the reference work and next stage.

The observed baseline is consistent with the reference work’s treatment of Pauli gates under an unbiased Pauli noise model: the paper reports substantial confusion among X , Y , and Z gate classes and describes them as sharing a non-distinctive fingerprint in that setting. The present experiment is narrower and is not a direct reproduction of the paper’s full setup; it tests only X_L versus Z_L using the installed **surface-sim** model and a Bernoulli Naive Bayes classifier.

The notebook therefore establishes a validated baseline for the next stage of SecureGate: implementing logical gates with more distinctive fault-tolerant signatures and evaluating whether detector spacetime patterns can support multi-class gate inference. The distinction between native backend operations and gates requiring explicit constructions will be preserved throughout that stage.

Reference. Shukla et al., “Anticipating Decoder Side-channel Attacks in Fault-tolerant Quantum Computers,” arXiv:2607.12174 (2026).